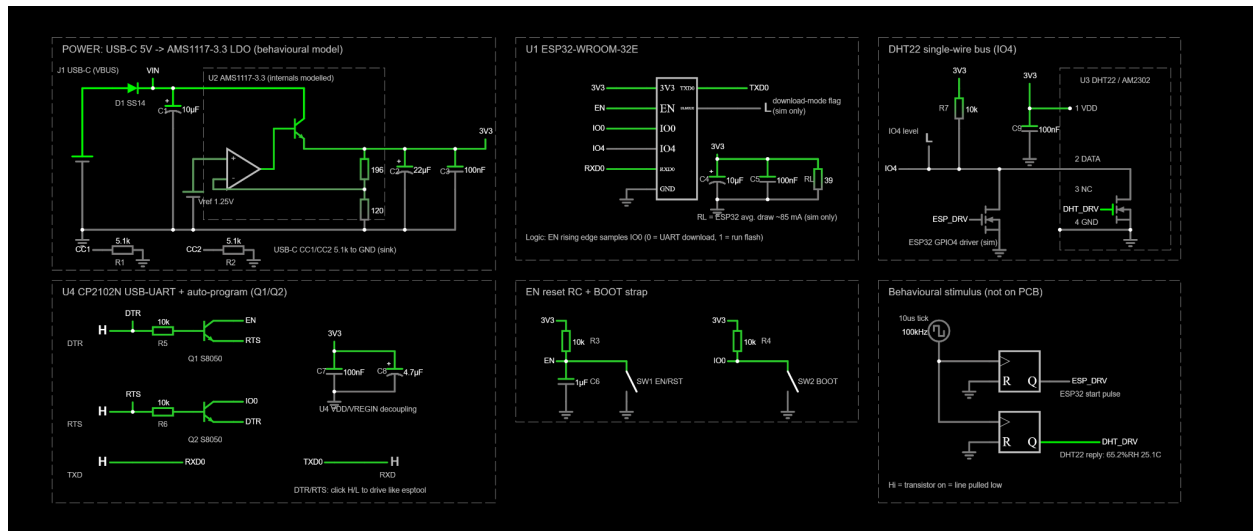


Part B, question 2



Component table

In	Qty	Value / Part	Function
J1	1	USB Type-C receptacle (USB 2.0)	5 V power input
R1, R2	2	5.1 kΩ	USB-C CC1/CC2 pull-downs
D1	1	SS14 Schottky diode	Reverse-current protection on VBUS
U2	1	AMS1117-3.3	3.3 V regulator
C1	1	10 µF	Regulator input capacitor
C2	1	22 µF tantalum	Regulator output capacitor
C3	1	100 nF	Regulator output decoupling
U1	1	ESP32-WROOM-32E	Microcontroller module
C4	1	10 µF	ESP32 bulk decoupling
C5	1	100 nF	ESP32 decoupling
R3	1	10 kΩ	EN pull-up
C6	1	1 µF	EN reset delay
SW1	1	Tactile switch	EN / reset button
R4	1	10 kΩ	GPIO0 pull-up
SW2	1	Tactile switch	BOOT button
U4	1	CP2102N-A02-GQFN28R	USB-to-UART bridge
C7	1	100 nF	CP2102N decoupling
C8	1	4.7 µF	CP2102N bulk decoupling
Q1, Q2	2	S8050 NPN transistor	Auto-program (DTR/RTS) circuit
R5, R6	2	10 kΩ	Q1/Q2 base resistors
U3	1	DHT22 / AM2302	Temperature and humidity sensor

R7	1	10 kΩ	DHT22 data pull-up
C9	1	100 nF	DHT22 decoupling
J2	1	2×8 pin header	16-pin I/O header with 5 V out
Total	27		